

How Much Do Traffic and Climate Explain Pavement Roughness?

Regression and classification models of the International Roughness Index on LTPP data, with location-grouped validation

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Abstract

Pavement roughness, measured by the International Roughness Index (IRI), drives ride quality, vehicle operating costs and maintenance decisions. This study asks how much of the IRI of in-service pavements can be explained by traffic loading, climate and section history, and how much depends on knowing a section's earlier condition. Using 15,935 annual IRI observations on 1,550 test sections from the Long-Term Pavement Performance (LTPP) programme (1989 to 2025), linear regression implemented with gradient descent, polynomial ridge regression, decision trees, random forests and logistic regression were compared against simple baselines. All models were evaluated with five-fold cross-validation grouped by location, so that sections sharing the same climate record never appeared in both training and test folds. Traffic, climate and section attributes explained little roughness (cross-validated $R^2 = 0.04$, RMSE = 0.575 m/km against 0.592 m/km for the mean). Adding the previously measured IRI raised R^2 to 0.93 (RMSE = 0.160 m/km), but a persistence forecast that simply repeats the last value already achieved $R^2 = 0.91$ (RMSE = 0.178 m/km); the remaining features reduced error by about 10% and explained only 5% of the change in IRI between surveys. The evaluation design proved decisive: a random forest without previous IRI scored $R^2 = 0.36$ when folds were grouped by section, 0.11 when grouped by LTPP site and -0.03 when grouped by location. For identifying poor pavements (IRI above 2.68 m/km), logistic regression with previous IRI caught 83% of poor observations at 88% precision, compared with 71% and 93% for persistence. The results indicate that previous condition dominates short-term roughness prediction, that traffic and climate without structural information carry little transferable signal, and that grouped validation is essential when evaluating machine-learning models on LTPP data.

Keywords: International Roughness Index; pavement performance; LTPP; machine learning; linear regression; random forest; logistic regression; cross-validation; data leakage

1. Introduction

The International Roughness Index (IRI) is the most widely used measure of pavement ride quality. It summarises the longitudinal profile of a wheel path as the accumulated suspension motion of a standard quarter-car model per unit distance travelled, expressed in m/km (Sayers, Gillespie and Paterson, 1986). Because IRI is linked to user comfort, vehicle operating costs and pavement deterioration, it is central to pavement management. In the United States, federal performance rules classify pavement sections as good when IRI is below 95 in/mi (1.50 m/km) and poor when it exceeds 170 in/mi (2.68 m/km) (FHWA, 2017).

Predicting IRI is therefore a core task of pavement management systems. Empirical roughness progression models, such as those in the World Bank's Highway Design and Maintenance model, predict the increment in roughness over time from the current roughness, traffic loading, pavement strength, age and environment (Paterson, 1987). The mechanistic-empirical design procedure adopted by AASHTO predicts IRI as an initial value plus contributions from distresses and a site factor that reflects climate and subgrade (AASHTO, 2008). Both approaches treat existing condition as the starting point and model deterioration from it.

More recently, machine-learning methods have been applied to LTPP data to predict IRI, and studies report high accuracy. Gong *et al.* (2018) used random forest regression to predict IRI of asphalt pavements, and Damirchilo *et al.* (2021) compared several machine-learning algorithms for IRI prediction using LTPP data. High reported accuracy raises two questions that motivate the present study. First, how much of the predictive skill comes from traffic and climate, and how much from information about a section's current or initial condition? Second, how sensitive are such results to the evaluation design? LTPP data are panel data: each section is surveyed repeatedly, and many sections are clustered at shared sites with common traffic and climate. When observations are correlated in this way, randomly or loosely split training and test sets can overstate performance (Roberts *et al.*, 2017), a form of data leakage (Kaufman *et al.*, 2012).

The aim of this study is to quantify how well traffic, climate and section history predict pavement roughness on LTPP sections, using an evaluation design that prevents leakage between related sections. The objectives are to:

- measure the predictive skill of traffic, climate and section attributes alone;
- measure the added value of previous condition relative to a persistence baseline;
- compare linear, polynomial, tree-based and logistic models under identical validation;
- test how the grouping used in cross-validation (by section, site or location) changes the conclusions; and
- assess whether the same information can identify pavements in poor condition.

2. Data

2.1 Source and extraction

Data were extracted from LTPP InfoPave (FHWA, 2026). IRI observations came from the ANALYSIS_IRI table; annual traffic estimates (equivalent single axle loads, ESAL; average annual daily truck traffic, AADTT; and truck volumes by FHWA vehicle class) from the TRF_TREND tables; construction events, pavement family and experiment type from EXPERIMENT_SECTION; functional class from SHRP_INFO; and annual climate (mean air temperature, freezing index and freeze-thaw cycles) from the MERRA-based climate tables linked to each section.

2.2 Processing

Repeated profiler runs on each visit were averaged and the survey year was taken from the visit date. Traffic and climate were joined to each section and year. Sections were identified by state code and SHRP_ID, since SHRP_ID values repeat across states. Duplicate rows (35), rows without traffic data (10) and rows with an impossible Class 9 share of trucks above 100% (5) were removed. Wheel-path IRI and mean roughness index columns were excluded because they are components or duplicates of the target. The final dataset contains **15,935 observations on 1,550 sections in 62 US states and Canadian provinces, surveyed between 1989 and 2025** (Table 1). Sections were surveyed a median of 10 times (range 1 to 27).

Table 1. Dataset summary

Statistic	Value
Observations	15,935
Sections	1,550
Locations (shared climate record)	471
LTPP sites (SPS site rule)	823
States and provinces	62
Survey years	1989 to 2025
IRI mean (SD)	1.35 (0.62) m/km
IRI median (range)	1.20 (0.32 to 5.74) m/km
Condition: good / fair / poor	70.8% / 24.9% / 4.4%

2.3 Features

Ten features describe each observation (Table 2). Traffic variables were log-transformed because of strong right skew. Age is measured from the start of the current construction event; for sections on their first construction number this is the date the section entered LTPP rather than the original construction date, which is a limitation discussed in Section 5.4. Two further features were derived for the second feature set: **PREV_IRI**, the IRI at the previous survey of the same section and construction event, and **GAP_YRS**, the years since that survey (median 1, range 1 to 9). Observations without a previous survey in the same construction event were excluded from the modelling set, leaving 11,581 observations on 1,527 sections and 470 locations.

Table 2. Features. Set A contains the first ten; set B adds PREV_IRI and GAP_YRS

Feature	Description	Mean	Min to max
AGE_SINCE_CN	Years since start of current construction event	4.6	0 to 29
IS_REHAB	1 if the section has been rehabilitated	0.70	0 / 1
IS_RIGID	1 if portland cement concrete surface	0.05	0 / 1
URBAN	1 if urban functional class	0.07	0 / 1
LOG_ESAL	$\ln(1 + \text{annual ESAL})$	11.7	0 to 15.3
LOG_AADTT	$\ln(1 + \text{average annual daily truck traffic})$	6.2	0.7 to 9.6
SHARE_CLASS9	Class 9 trucks as share of all trucks	0.47	0 to 0.89
TEMP_AVG	Mean annual air temperature (°C)	11.5	-5.4 to 26.2
FREEZE_INDEX	Annual freezing index (°C-days)	384	0 to 3,647
FREEZE_THAW	Annual freeze-thaw cycles	80	0 to 189
PREV_IRI (set B)	IRI at previous survey (m/km)	1.27	0.32 to 5.74
GAP_YRS (set B)	Years since previous survey	1.5	1 to 9

3. Methods

3.1 Evaluation design

Three structures in LTPP data create dependence between observations. Repeated surveys of a section are strongly correlated: section identity alone accounts for 61.7% of the variance in IRI. Specific Pavement Studies (SPS) sections are built side by side at one site and share traffic and climate. More generally, sections that share a climate grid cell have identical climate histories. To prevent models from recognising a location seen during training, observations were grouped into **locations**: sections that share any identical annual climate record (same state, year, temperature, freezing index and freeze-thaw count) were joined into one group, which yields 471 locations and places all SPS sections of a site together. All models were compared with five-fold GroupKFold cross-validation on these locations, so every location appears in exactly one test fold (Roberts *et al.*, 2017). To show the effect of the grouping choice, key models were also cross-validated with folds grouped by section and by SPS site. Classification results and figures use a single held-out split of 20% of locations (2,844 observations, 94 locations).

3.2 Baselines

Three baselines anchor the comparisons: the mean IRI of the training folds; **persistence**, which predicts that IRI equals the previous survey's value; and, for classification, a rule that labels every observation as not poor.

3.3 Models

- **Linear regression.** Implemented from first principles with batch gradient descent on standardised features (learning rate 0.1, 1,000 iterations) and verified against the closed-form scikit-learn solution (Pedregosa *et al.*, 2011); the coefficients agreed to four decimal places.
- **Polynomial ridge regression.** Polynomial expansions of degree 1 to 4 with an L2 penalty (Hoerl and Kennard, 1970), with the penalty strength λ chosen by grouped cross-validation.
- **Decision trees and random forests.** Random forests (Breiman, 2001) of 300 trees with a minimum of five observations per leaf.
- **Logistic regression.** For the binary outcome poor (IRI above 2.68 m/km), implemented with gradient descent and verified against scikit-learn, with analysis of decision thresholds and balanced class weights.

Each regression model was fitted with feature set A (traffic, climate and section attributes) and feature set B (set A plus previous IRI and survey gap). Set B corresponds to the incremental form used in roughness progression models (Paterson, 1987).

3.4 Metrics

Regression performance was measured with the coefficient of determination (R^2), root mean squared error (RMSE) and mean absolute error (MAE), averaged across folds. Classification was measured with precision, recall and F1 for the poor class, since about 4% of observations are poor and accuracy is uninformative under such imbalance. Feature influence was assessed from standardised coefficients and from permutation importance on the held-out locations (Breiman, 2001).

4. Results

4.1 Exploratory findings

IRI was strongly right-skewed (skewness 1.71). A log transformation reduced the skewness to 0.35 but gave a lower test R^2 after back-transformation (0.004 against 0.068 for the untransformed target in the guided section split), so IRI was modelled on its original scale (Figure 1). No single feature had a strong monotonic association with IRI: absolute Spearman correlations ranged from 0.01 to 0.23, the largest being for rigid pavement (0.23), which had a median IRI of 1.84 m/km against 1.17 m/km for flexible pavement. Traffic was weakly negatively associated with IRI (-0.12 for AADTT), consistent with heavier-trafficked routes receiving thicker designs and more maintenance.

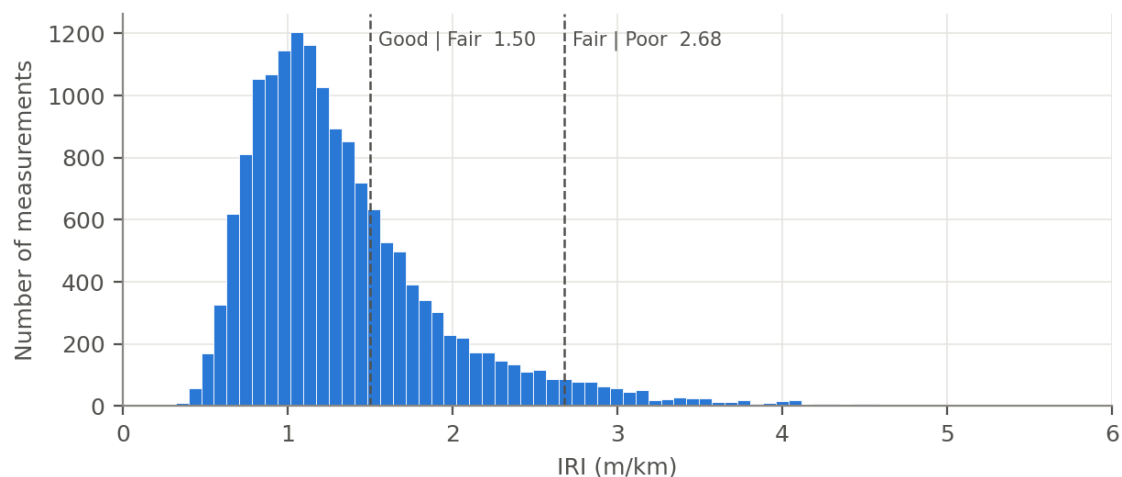


Figure 1. Distribution of IRI with the FHWA good/fair and fair/poor thresholds.

Median IRI was almost unchanged with age across all sections, varying only between 1.13 and 1.25 m/km for ages 0 to 15 years (Figure 2a). Within sections, however, IRI rose steadily, by a median of about 0.2 m/km after 15 years relative to each section's first survey (Figure 2b). The flat cross-sectional pattern reflects rehabilitation of rough sections, which resets age, and large differences in initial roughness between sections.

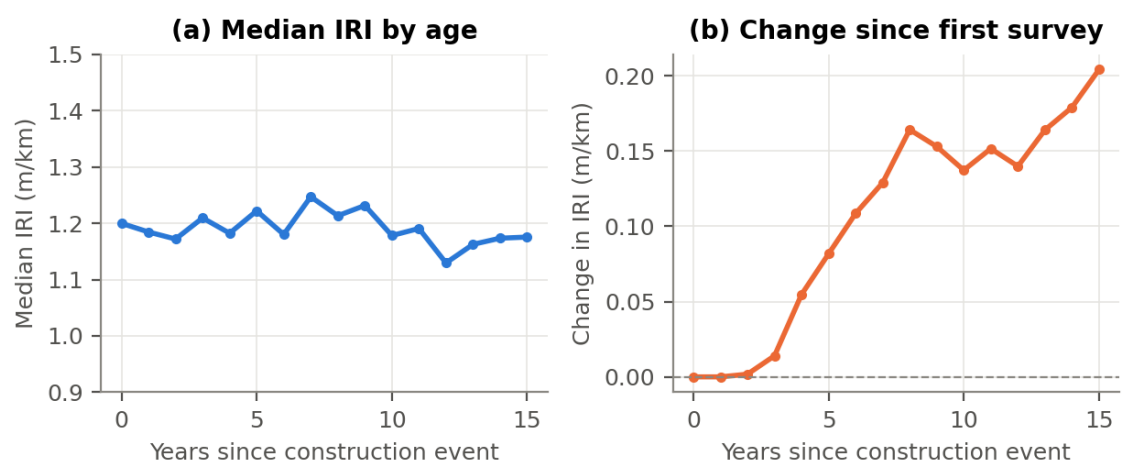


Figure 2. (a) Median IRI by age across all sections. (b) Median change in IRI since each section's first survey. Ages with fewer than 100 observations are omitted.

4.2 Regression performance

Table 3 summarises the location-grouped cross-validation. With feature set A, linear regression explained 4.4% of the variance (RMSE 0.575 m/km), only slightly better than predicting the training mean (RMSE 0.592 m/km). Polynomial ridge regression gave no further gain, and the random forest performed worse than the mean. With feature set B, linear regression reached $R^2 = 0.925$ (RMSE 0.160 m/km) and the random forest $R^2 = 0.922$. The persistence baseline alone achieved $R^2 = 0.908$ (RMSE 0.178 m/km). The learned models therefore reduced RMSE by about 10% relative to persistence. When the same set B model was used to predict the change in IRI since the previous survey, it explained 5.1% of that change.

Table 3. Five-fold cross-validation grouped by location (11,581 observations, 470 locations)

Model	Features	R^2 (SD)	RMSE (m/km)	MAE (m/km)
Mean of training folds	none	-0.016 (0.010)	0.592	0.435
Linear regression	A	0.044 (0.063)	0.575	0.421
Polynomial (degree 2) + ridge	A	0.045 (0.082)	0.575	0.422
Random forest	A	-0.032 (0.062)	0.597	0.448
Persistence	previous IRI	0.908 (0.011)	0.178	0.099
Linear regression	B	0.925 (0.010)	0.160	0.093
Random forest	B	0.922 (0.010)	0.164	0.097

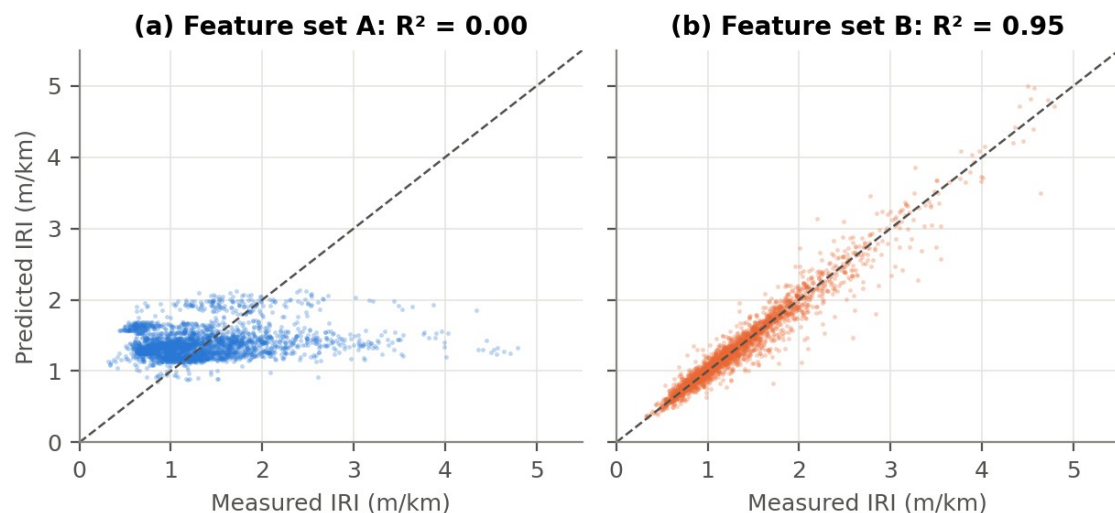


Figure 3. Predicted against measured IRI on held-out locations for linear regression with (a) feature set A and (b) feature set B.

Figure 3 illustrates the difference. With set A the predictions are compressed into a narrow band around the mean, and the model cannot separate rough sections from smooth ones. With set B the predictions follow the measurements closely, largely because they track the previous value.

4.3 Model complexity

Increasing polynomial degree on set A raised training R^2 from 0.10 (degree 1) to 0.38 (degree 4) while the median test-fold R^2 fell from 0.077 to -5.7, with the worst folds far below zero (Table 4). Ridge regularisation restrained the degree-3 model, but its best cross-validated R^2 (0.043 at $\lambda = 10,000$) did not exceed that of the straight-line model. Additional flexibility did not compensate for the limited information in the features.

Table 4. Polynomial degree and overfitting, feature set A (all 15,935 observations, location-grouped folds)

Degree	Features	Training R^2 (mean)	Test R^2 (median fold)	Test R^2 (worst fold)
1	10	0.101	0.077	0.004
2	65	0.159	0.045	-0.024
3	285	0.257	-0.293	-13.4
4	1,000	0.378	-5.69	-14,966

4.4 Sensitivity to the validation grouping

The grouping used for cross-validation changed the conclusions for flexible models (Table 5). A random forest with feature set A appeared to explain 36% of the variance when folds were grouped by section, 11% when grouped by SPS site and -3% when grouped by location. Linear regression was much less affected (0.087, 0.033 and 0.044), and models with previous IRI were almost unaffected. The forest's apparent skill under section grouping therefore came mainly from recognising the traffic and climate signature of locations already seen in training.

Table 5. Cross-validated R^2 under three grouping schemes

Model	Grouped by section	Grouped by SPS site	Grouped by location
Linear regression, set A	0.087	0.033	0.044
Random forest, set A	0.360	0.113	-0.032
Linear regression, set B	0.927	0.925	0.925
Random forest, set B	0.927	0.921	0.922

4.5 Influential factors

In the set B linear model with standardised features, previous IRI had by far the largest coefficient (0.562 m/km per standard deviation), followed by the survey gap (0.033), freezing index (0.021) and mean temperature (0.013); all other coefficients were below 0.01 in magnitude. Permutation importance for the set B random forest gave the same ranking: shuffling previous IRI reduced R^2 by 1.86, and among the remaining features the survey gap was the most important (Figure 4). In the set A linear model, rigid pavement, temperature and freezing index had the largest positive coefficients, while the two traffic measures had opposite signs (LOG_ESAL +0.05, LOG_AADTT -0.10), a symptom of their strong collinearity.

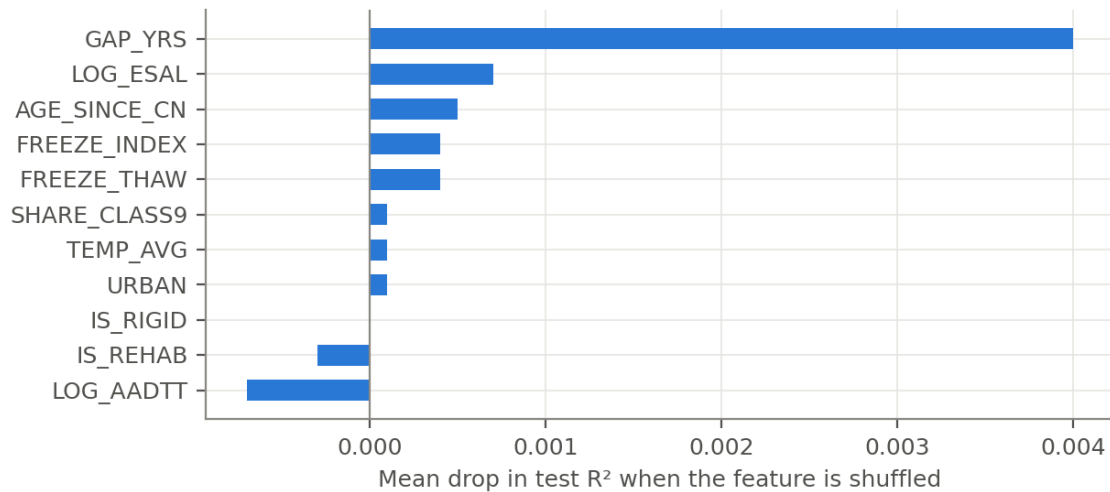


Figure 4. Permutation importance for the random forest with feature set B on held-out locations, excluding previous IRI (1.86).

4.6 Classifying poor pavements

On the held-out locations, 107 of 2,844 observations (3.8%) were poor (Table 6). Labelling every observation as not poor achieved 96% accuracy while detecting none. Persistence detected 71% of poor observations with 93% precision. Logistic regression with set B at the default threshold of 0.5 detected 83% with 88% precision ($F1 = 0.86$). Lowering the threshold increased recall at the expense of precision (Figure 5): at 0.1 it detected 96% of poor observations with 52% precision. Without previous IRI, a class-balanced model detected 63% of poor observations but only 6% of its alerts were correct.

Table 6. Classification of poor condition on held-out locations (107 poor of 2,844 observations)

Approach	Precision	Recall	F1	Missed	False alarms
Always not poor	0.000	0.000	0.000	107	0
Persistence (previous IRI above 2.68)	0.927	0.710	0.804	31	6
Logistic, set B, threshold 0.5	0.881	0.832	0.856	18	12
Logistic, set B, threshold 0.3	0.744	0.897	0.814	11	33
Logistic, set B, threshold 0.1	0.515	0.963	0.671	4	97
Logistic, set B, balanced weights	0.369	0.991	0.538	1	181
Logistic, set A, balanced weights	0.056	0.626	0.103	40	1,132

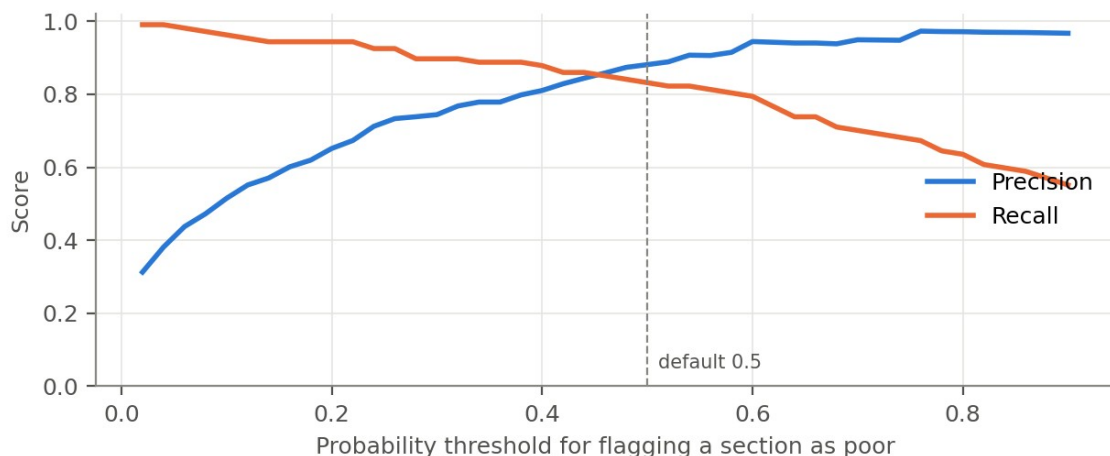


Figure 5. Precision and recall of the set B logistic model as the decision threshold changes.

5. Discussion

5.1 Traffic and climate alone carry little transferable signal

Across linear, polynomial and tree-based models, traffic, climate and section attributes explained at most about 4% of the variance in IRI on unseen locations. Three explanations are consistent with the data. First, the feature set omits pavement structure and materials (layer thicknesses, base type, asphalt properties), which govern how a pavement responds to loading and environment and which form the core of mechanistic-empirical models (AASHTO, 2008). Second, traffic is confounded with design and maintenance: heavily trafficked routes are built thicker and maintained more often, which explains the weak negative association between truck volume and IRI. Third, most variation in IRI lies between sections (61.7%), reflecting as-built smoothness and structural differences that the features do not describe.

5.2 Previous condition dominates

Adding previous IRI raised R^2 from 0.04 to 0.93, which agrees with the incremental structure of established roughness progression models (Paterson, 1987). However, comparison with persistence places this result in perspective. Repeating the previous value already explains 91% of the variance, so the learned models add about two percentage points of R^2 and a 10% reduction in RMSE, and they explain only about 5% of the change in IRI between surveys. The variables that contributed to that change, namely time since the previous survey, freezing index and temperature, are physically plausible: deterioration accumulates with time, and frost action is a recognised driver of roughness. The practical implication is that high R^2 values for IRI prediction should always be reported alongside a persistence baseline, since a model can appear highly accurate while contributing little beyond the most recent measurement.

5.3 Validation design matters

The most striking methodological result is the sensitivity of the random forest to the grouping of cross-validation folds. Grouping by section, which already prevents the same section from appearing in training and test folds, still allowed the forest to exploit shared traffic and climate among neighbouring sections, and its apparent R^2 of 0.36 fell to about zero when whole locations were held out. This is a form of leakage in which the test data share information with the training data through structure rather than through duplicated rows (Kaufman *et al.*, 2012), and it is the situation for which grouped or blocked cross-validation is recommended (Roberts *et al.*, 2017). Flexible models are more exposed to this leakage than linear models because they can memorise specific combinations of feature values. Reported accuracies from machine-learning studies on LTPP data depend on the features used and on how data are split, so comparisons between studies should take both into account.

5.4 Limitations

- **Age definition.** For first construction events, age counts from entry into LTPP rather than original construction. True construction dates (table INV_AGE) were not extracted.
- **Missing variables.** Layer thicknesses, material properties, precipitation, subgrade and detailed maintenance history were not included.
- **Traffic data.** Annual traffic values are trend estimates, not direct measurements for every year.
- **Location grouping.** Locations were inferred from identical climate records, which is conservative but approximate.
- **Representativeness.** LTPP sections are monitored and maintained more closely than typical network sections, so results may not transfer directly to other networks, including those in West Africa.

- **Single threshold.** Classification used one condition threshold; results for other thresholds or for fair condition were not examined.

6. Conclusions and future work

This study compared regression and classification models of pavement roughness on 15,935 LTPP observations using location-grouped cross-validation. The main conclusions are:

- Traffic, climate and section attributes without structural information explained only about 4% of the variance in IRI on unseen locations, and no model type, linear, polynomial or tree-based, improved on this.
- Previous condition dominated prediction. The best model (linear regression with previous IRI and survey gap, $R^2 = 0.925$, RMSE = 0.160 m/km) improved on a persistence forecast ($R^2 = 0.908$, RMSE = 0.178 m/km) by about 10% in RMSE, with survey gap, freezing index and temperature as the most useful additional variables.
- The validation grouping changed the conclusions. A random forest's apparent R^2 fell from 0.36 to about zero when whole locations, rather than sections, were held out.
- For identifying poor pavements, logistic regression with previous IRI detected 83% of poor observations at 88% precision, and the decision threshold offers an explicit trade-off between missed poor sections and false alarms.

Future work should add true construction age, pavement structure and material data, and precipitation; model section-specific deterioration rates with mixed-effects or hierarchical models; evaluate gradient boosting under the same grouped validation; and apply the framework to road condition data from Ghana and other West African networks, where calibrated roughness progression models are needed for maintenance planning.

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References

- AASHTO (2008) *Mechanistic-Empirical Pavement Design Guide: A Manual of Practice*. Interim edn. Washington, DC: American Association of State Highway and Transportation Officials.
- Breiman, L. (2001) 'Random forests', *Machine Learning*, 45(1), pp. 5–32.
- Damirchilo, F., Hosseini, A., Mellat Parast, M. and Fini, E.H. (2021) 'Machine learning approach to predict international roughness index using Long-Term Pavement Performance data', *Journal of Transportation Engineering, Part B: Pavements*, 147(4).
- FHWA (2017) 'National performance management measures; assessing pavement condition for the National Highway Performance Program and bridge condition for the National Highway Performance Program', final rule, 23 CFR Part 490, *Federal Register*, 82(11).
- FHWA (2026) *LTPP InfoPave*. Federal Highway Administration. Available at: <https://infopave.fhwa.dot.gov> (Accessed: 26 September 2026).
- Gong, H., Sun, Y., Shu, X. and Huang, B. (2018) 'Use of random forests regression for predicting IRI of asphalt pavements', *Construction and Building Materials*, 189, pp. 890–897.
- Hoerl, A.E. and Kennard, R.W. (1970) 'Ridge regression: biased estimation for nonorthogonal problems', *Technometrics*, 12(1), pp. 55–67.

- Kaufman, S., Rosset, S., Perlich, C. and Stitelman, O. (2012) 'Leakage in data mining: formulation, detection, and avoidance', *ACM Transactions on Knowledge Discovery from Data*, 6(4), article 15.
- Paterson, W.D.O. (1987) *Road Deterioration and Maintenance Effects: Models for Planning and Management*. Baltimore: Johns Hopkins University Press for the World Bank.
- Pedregosa, F. *et al.* (2011) 'Scikit-learn: machine learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830.
- Roberts, D.R. *et al.* (2017) 'Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure', *Ecography*, 40(8), pp. 913–929.
- Sayers, M.W., Gillespie, T.D. and Paterson, W.D.O. (1986) *Guidelines for Conducting and Calibrating Road Roughness Measurements*. World Bank Technical Paper No. 46. Washington, DC: World Bank.